Practical 2: Data Preprocessing - Submission Template

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Practical Number: 2 - Data Preprocessing
IDE Used: ☐ VS Code ☐ Google Colab

✅ Learning Outcomes Verification

LO1: Identified data quality issues
LO2: Handled missing values effectively
LO3: Detected and treated outliers
LO4: Scaled and normalized features
LO5: Encoded categorical variables
LO6: Created reusable pipelines
LO7: Validated preprocessing quality
LO8: Documented all steps

📊 Preprocessing Statistics

Original Dataset:

- Rows: 150
- Columns: 6
- Missing Values: 5
- Duplicates: 2

After Cleaning:

- Rows: 148
- Missing Values: 0
- Duplicates: 0
- Outliers Removed: 3

Final Dataset:

- Rows: 145
- Columns: 8 (after encoding)
- Data Types: All Numeric (✓)

🧪 Test Results

Test 1: Data Loading

Status: ✓ PASSED
Output: Dataset loaded successfully using Pandas DataFrame.

Test 2: Missing Value Handling

Strategy Used: Mean Imputation / Deletion / Forward Fill
Status: ✓ PASSED
Missing Values Before: 5
Missing Values After: 0

Test 3: Outlier Detection

Method Used: IQR

Outliers Found: 3

Outliers Removed: 3

Status: ✓ PASSED

Test 4: Feature Scaling

Scaler Used: StandardScaler

Before Mean: 45.62 | After Mean: 0.00

Before Std: 12.18 | After Std: 1.00

Status: ✓ PASSED

Test 5: Categorical Encoding

Encoding Method: One-Hot Encoding

Categories Encoded: Gender, City

New Features Created: 4

Status: ✓ PASSED

📁 Files Submitted

- `Practical_2_Complete_Notebook.ipynb`
- `SUBMISSION_TEMPLATE_Practical_2.md`
- Cleaned dataset (CSV)
- Screenshots (3-4)

Submission Status: ✓ Complete